QUEENSLAND UNIVERSITY OF TECHNOLOGY

EGH490-2 · Research Project 2

PROJECT DELIVERY REPORT

Controlling Differential Power Processors in Solar Powered Applications

THE DESIGN AND IMPLEMENTATION OF MODERN STATE-SPACE METHODS TO SERIAL PARTIAL POWER CONVERTERS

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1.0	23/10/26	Jared Saunders	Project Delivery Report

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1 Executive Summary

The device constructed in this report is a controller for a serial partial power converter that has been coined a 'Power Shuffler'. This type of converter is a part of the family, differential power processors, whose reason for being is to move differences in power around a system, effectively balancing out either voltage or current. The serial partial power converter specialises in moving differentials of modules that are connected in series. There is a sister report to this one that covers in-depth the electrical design of the power shuffler, whereas the content of this report is focused on controlling it. The target application is solar panels. One of the biggest limitations of solar panels as a technology is their massive performance drop in shade. Normally solar panels are connected in series which means they share the same current but the voltage increases as they are stacked on top of each other. When a panel is shaded it reverse-biases and limits the current that can flow in the string. The most severely shaded one becomes the weakest link in the chain. Solar panel manufacturers mitigate this by implementing bypass diodes that as the name suggests skips groups 1/3 or 1/2 of a panel. Thereby, foregoing the power that could have been collected by that group.

A single shuffler decouples two pv modules from the string using capacitors and creates a bidirectional connection using a coupled inductor and two half-bridges. This type of s-ppc is called an interleaved buck-boost converter. A normal buck-boost can be controlled by constructing by implementing classical methods on a single 2nd order ordinary differential equation. However, the interleaved buck-boost results in multiple ODEs. Classical methods will not help us here. A collection of single input single output transfer functions cannot accurately describe the dynamics of the system. Therefore, the modern control method of state space modelling captures all the relationships between the states with its multiple input multiple output capabilities. A simplified state-space model was determined to act as the basis for controller designs. As the project progressed the analytical model was updated to retro-actively as the system developed. The states selected were the phase currents (ie. currents that pass through a particular phase leg to the inductor) and pv module voltages. These grant controllability and observability of the system.

Based on these insights the model was adapted to simulink as a high-level controller architecture. There were three simulink files created. The first, Model In The Loop is a pure simulink adaptation of the average switching model. At this point I was able to select sensors based on observability requirements and select a microcontroller from controllability requirements. Hall Effect Sensor TMCS11 for phase currents. Power monitor INA239 senses module voltages. The STM32 NUCLEOG474RE is the microcontroller be suited for the project with its large number of peripherals, fast clock speeds and high resolution ADC. For this project the minimum viable product features required for controller implementation must include two shufflers and a microcontroller. The microcontroller was to be bought separately and mounted to the PCB by headers instead of placing just the chip. The signals routed from the headers of the micro include a 3v3 rail, nucleo GND rail, SPI bus for voltage sensing, ADC lines for current sensing and PWM channels for gate driver control. Kicad was used for the PCB design.

The manufacturing process. PCBway board / stencil order. Components order. Solder paste and bake in a pcb oven.

The pipeline for generating new gains. SIL SIMULINK (Coarse Gains) –> QSPICE Integrated Model (Refined Gains + Noise Injection) –> RTOS (Implemented in Firmware).

Extra features were added to make it a self-contained system. These include an enclosure, heatsink, microcomputer, remote control, message screen and a status LED. The enclosure protects the shuffler, attaches the heatsink and encapsulates / mounts all parts of the system. The raspberry pi 4B is primaraily a data logger, remote flasher to the the nucleo, user interface for the controller, power distribution unit for these accessories and the nucleo. The features also improve accessibility and usability.

Bench Testing was a success. The setup was a 3-channel power supply (NGE103), an electronic load (DSB3021), oscilloscope, laptop and the full system. Started at 200kHz open loop to get some baseline behaviour. Then closed loop with gains of (0.1, 0.2, 0.3) of QSPICE values for pv panel setup. Then we progressed to 300 kHz at (0.3 and 0.5). Then to 400 kHz at 0.3. 300 kHz seemed to be the most optimal. There appears to be a trade-off where the benefits reduce as the switching speed increases and the heat increases. We also explored an open loop test on 100 kHz which the system has sort of working but bizarre behaviour to. It is concluded that the power supplies can not handle as much as the panels. Or do not require as aggressive control.

Limitations: Can only do 2 shufflers. Boards can not be stacked together. Can not comment on system scalability. The shufflers struggle to be driven independently as they are hardwired together. The top-level ina239 does not function had to guess based on eload value. Have to rerun the sim everytime the switching speed changes for best results. Did not test on solar panels so response to panel dynamics is only theoretical.

Future work: Scalability. Determine how the relationships change when more shufflers are acting in a string or how a shuffler handles varying number of panels. Streamline switching speed gain calcs. Generate and compare control models. Apply a machine learning model / cost function to determine optimal gains. Fix the top-level voltage sensor. Consider effectiveness with different panel arrangements. How does it stack up against dynamic panel configurations? Explore other applications outside the solar industry. Think about how it applies to other sectors of the energy industry like mining, computing, industrial, batteries etc. Would this let you win a solar car race? Is that game changing? Does the technology itself scale?

Recommendations: Explore other contexts. Beware of the electrical safety limits. They do restrict the experiments you can actually run. Go for hand on development for understanding The system likes high switching frequencies, possibly try revisiting GAN. Mechanical considerations: downsize the heatsink, improve fasteners, pcb reconfigure, redesign enclosure. Review our github, manufacturing notes and user guides for details on how to setup your own project and continue it. Consolidate the microcontroller and the microcomputer into one board.

2 Abstract

Intro / Background :
Solar panels are the highest yield renewable resource available to the world. They cover 10% of Australia's energy production. When panel of a solar array becomes shaded it proportionally limits the current of the array and eventually reverse-biases blocking power to the rest. Industry tends to bypass the portions or whole solar panels with bypass diodes however, in doing so it leaves untapped energy on the table as part of the trade-off. In an effort to extract this energy a device a device that has been named the 'Power Shuffler' has been constructed that works to equalise the voltage in this system by moving current up and down the array. It does this by establishing a bidirectional connection with an interleaved buck-boost converter architecture.

Purpose / Objectives : To track the maximum power point of a partially shaded photovoltaic system using this technology a controller must be developed. The complexity of the dynamics of the converter requires a modern state space control system that can handle the multiple input multiple output nature. The objective is to design and implement a controller that integrates with the converter and a PV system to extract extra power.

Methodology : Methodology style was systems engineering life-cycle approach. The development of this controller starts began with determining an analytical solution, and move into a high-level simulink controller architecture, then printed circuit board development, combining controller and hardware simulations in QSPICE and finally developing a Real-Time-Operating-System to bring the controller to life on the bench and out and in the sun.

Results / Key Findings : We were able to extract up to 25% more power than would otherwise would be possible. Currents were constrained to plus-minus 200mA, as well as module voltages +- 200mV for a gain of 30% QSPICE at 300 KHz during lab testing. 400 kHz showed marginal efficiency gains. 200 kHz is the lower limit of the system.

Conclusion / Implications : The controller for the power shuffler succeeds at extracting extra power during partial shading events, is stable, has limited overshoot, rises in 0.5ms, settles within 2 ms and has a steady state error of less than 1%. Next steps are to scale up the system to handle more than 3 PV modules in a string.

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3 Introduction

Problem Statement :
The maximum power point of a photovoltaic system is what most solar controllers track to get the most energy. When a PV string is partially shaded the MPP drops and multiple local maxima are created. The power shuffler addresses this problem by pinning a whole string to one MPP by creating new paths for currents to move up and down the string. Specifically, the MPP is found when all PV module voltages are equal.

Research Question :
How to develop a controller that captures the dynamics of multiple interleaved buck-boost converters and commands the currents in a way that extracts the maximum power point despite shading.

Aims :
The aim of this project is to design and implement a controller for the power shuffler.

Objectives :

- Derive an analytical solution to the dynamics of an interleaved buck-boost converter
- Design a control system architecture.
- Develop a controller board
- Experimental validation on a real photovoltaic system.
- Create a self-contained product to showcase controller autonomy.

4 Literature Review

Literature review: synthesis of previous research related to the topic and methodology of your research project. Identifying the gaps which your project has addressed in one paragraph and as a subsection

Partial-shading and MPPT

Solar controller = MPPT. This is typically done with buck/boost converters. Aim of MPPT is to get the highest efficiency power conversion. Partial shading directly challenges the MPPT. (illustrate MPP changes)

Existing Shuffler Concepts DPP / S-PPC

This type of converter is a part of the family, differential power processors [1], whose reason for being is to move differences in power around a system, effectively balancing out either voltage or current. The serial partial power converter specialises in moving differentials of modules that are connected in series [4]. Other researchers use DPP. For example (Jeong) created 4 DPP experiment. Explain how they don't dicuss their control algorithms or they keep it to just PI. They like to keep it simple and use classical.
The architetcure for the S-PPC that this shuffler uses an interleaved buck-boost converter. Basic functional explanation. (draw the generic schematic)

Controlling Interleaved Buck-Boost

Talk about the derivation for buck-boost and how it interleaved add complexity with more states that are 'coupled' are not controlled independently. [5], [6]. One of the impleemnted an extended kalman filter into their design. Talk about the PV state space model article. Describe the type of ODE. [2] This ties segues nicely to the gap of tieing all these concepts together.

The Gap:

Controllers in the past have been able to use classical control design methods such as PI. MIMO control has very limited exploration. Interleaved buck-boost converter control is very niche.

5 Methodology

Methodology: describe the methods used to address the research questions and why these are appropriate to your project, outline any stakeholder engagement, design, simulation, data collection or testing approaches used and interpretation, validation, and analysis techniques. Modelling work requires a detailed description of assumptions. Include any risk, ethics, and sustainability considerations which you accounted for in your research project.

5.1 Controller Design Pipeline

The methods applied to the project correspond to different stages of the v-model as seen in Figure 1. It is how systems engineers prefer to map out the development throughout a products lifecycle. As a result of the literature review the research problem has been identified and the research question articulated. The first step in the design process is to determine the target use-case and application of the product through background research resulting in clear objectives. To determine a set of system requirements for a controller the plant is modelled. Then the controller algorithms are modelled in simulink. Based on this abstract model components can be selected for a controller pcb. Knowing the controller architetcure especially the microcontroller and sensors opens the door for firmware development. The hardware and controller SIL simaulations can be combined together to provide a theoretical approximation of the experiments. Testing the system in the electronics lab. Testing outdoors with a string of 3 PV modules is the core objective and ultimate validation of the system. Table 1 outlines the connections between the overarching methodology for this project and conventional industry standards.

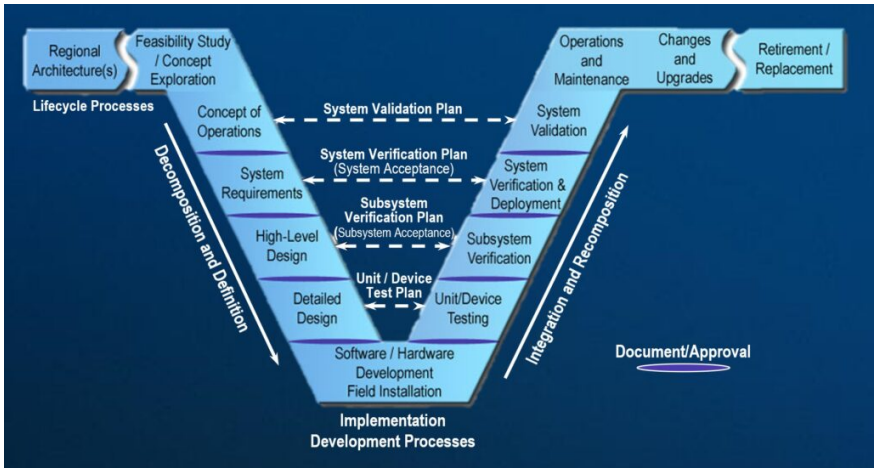


Figure 1: Systems Engineering Product Lifecycle, [3]

5.2 Background

- show the interleaved buck boost converter.
- describe its basic function.
- talk about what we are targeting. 2 shufflers, one controller, telemetry system, user interface, the foundation for sclaing, 3 PV modules in a string.
- explain why we rated the early designs to the highest rated solar panels on the commercial market. (AIKO 555)

Phase	Pipeline	V-Model
0	Background Research	Concept of Operations
1	Analytical Solution	System Requirements
2	Model-in-the-Loop	High-Level Design
3	Component Selection	Detailed Design
4	Firmware / Software	Development
6	Theoretical Integration	Subsystem Verification
7	Bench Test	System Verification
8	Solar Test Partial Shade	System Validation

Table 1: Development Phases Breakdown

5.3 Analytical Solution

- Create a drawing that captures the N-Modules problem. Should be easy to spot key physical parameters from the model in the drawing.
- derive the state space model for a single shuffler's dynamics. Quickly provide the basic stability, controllability, observability and equillibrium analysis
- derive the model for N-Modules. show how not only the dynamics change due to coupling but so does equilibrium, controllability, observability, stability.
- explain the difference between common mode and differential mode (might be better to refer to them as current control and voltage control modes)
- Show the DC gain identification
- should I put phase portraits, step response and root locus here or move them to a later section?

5.4 Model-in-the-Loop (MIL)

- show the model
- generate the pseudocode for the controller, supervisor, MPPT (keep it simple)
- show the plots that validate MIL
- talk about how SIL is simply an extension of MIL where it opens the door for C programs to be called instead of matlab functions, this is a cornerstone of the gain refinement pipeline.

5.5 Component Selection

- Talk about the high-level control data flow diagram.
- create a very detailed system architecture, (unsure whether I want it as one big web or a hierarchical set of diagrams). Maybe even show how the major functions / script interact. Maybe even a state machine.

- explain that the microcontroller was chosen for its veyr high clock speed, affordable price, prioritises adc functionality with up to 5 ADCs. High resolution adc and timers. Many timers in fact. ADC and PWM fucntionality were the driving factors.
- explain that the only reason the ina239 was chosen was because the ina229 was out of stock. The 229 had the better adc resolution. The inas were desirable as they could use the same spi bus.
- explain that a raspberry pi is the perfect and most safe to handle telemetry and other extra features without adding complexity to the pcb. the pi acts the user interface to the whole shuffler, allowing for remote control, state display, internet connectivioty, extra processing, can acts as a power distribution for 5v or 3v3, can flash the nucleo. The users laptop does not need to plug directly into the system. It is good enough to host the web services.
- explain that the hall effect sensor is a very safe, and high precision current sensor.
- defend my routing choices and show the diagram. (maybe even a kicad drawing). ADC needs to have the most directv the path and it was viad to the bottom to be as far away from switching waveforms as possible. SPI forms a bus going up and down. the pwm and chip selects were just whatever was convenient.
- add a sumamry table of the peripherals (relates to the header pinout). Used stm32cubemx to help determine these.

5.6 Gain Refinement Pipeline

- Updated the simulink to better reflect the experiment for bench and solar. Changed the power sources, load, and reduced the plant to 3. added a sensing block to inject noise and delays that reflect teh sensors.
- integrated the control architecture into teh hardware guy's qspice (his name is Son Ngyugen)
- the qspice added a controller component to represent the nucleo (and its firmware on the dlls). Represented the sensor hardware too.
- Show the gain values produced by the simulink and compare them to the qspice.
- show pictures of the various layers of the simulink and qspice.
- how to run the sims and change the switching frequency.
- explain how simulink, qspice , and firmware share scripts.

5.7 Firmware / Software

- what firmware was created?
- is there a freeRTOS scheduler. what does it prioritise / what are the tasks?
- what are teh key configuration parameters for the perihperals. I'm talking speeds, trigger events and addresses and calibration. For instance I know that adc reads are triggered on the pwm edge I believ that is for comb filtering.
- basic pseudocode of important functions like the scheduler.

- talk about how the pi host grafana and influx. discuss docker and deployment.
- firmware / software validation for physical signals can be probed with an oscilloscope. Definitely done for adc and pwm signals. (can show a picture)
- Experimental setup
- what are we testing / looking for
- differences between bench test and solar test

5.8 Risks

Risk register table. Think about safety, people, faults at various technical stages, scope creep.

5.9 Ethics

5.10 Sustainability

6 Results, Analysis and Discussion

Results, Analysis, and Discussion: present the outcomes including any stakeholder feedback, engineering designs, simulation outcomes, data analysis, and performance data. Discuss these in relation to the gaps identified in the literature review and explain their significance. Briefly address the outcomes of your consideration of risk, ethics, and sustainability. Identify any limitations in your study

6.1 Proposed Solution

Figure 2 is a CAD render of the complete power shuffler device.

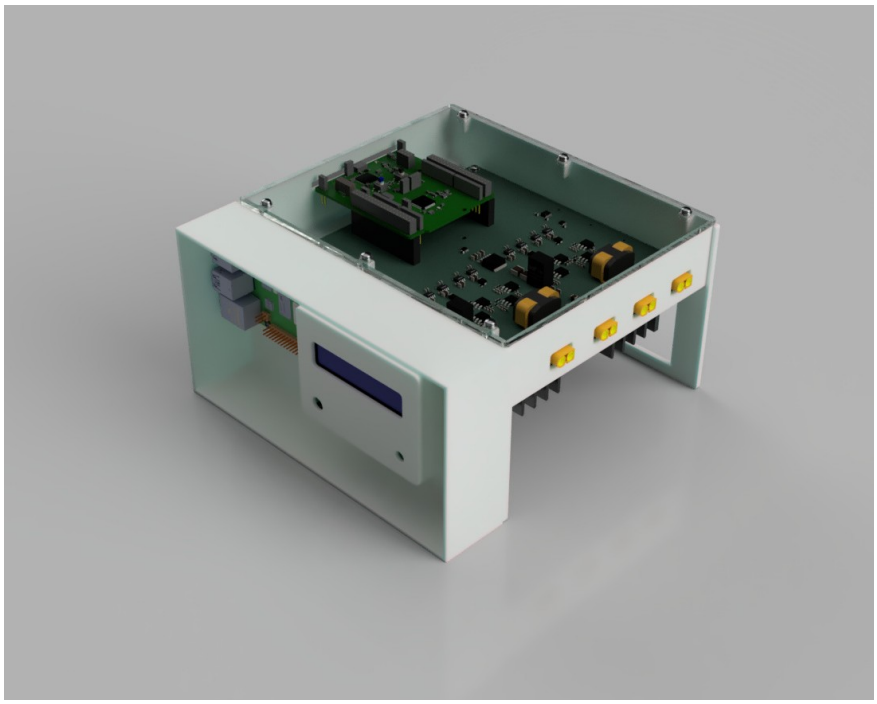


Figure 2: Power Shuffler Assembled

6.2 Analytical

talk about the findings. show phase portaits and root locus.

6.3 Simulink

Summarise the important simulink data. probably don't need to show MIL here it is implied by carpark working.

6.4 QSPICE

show the raw qspice data.

6.5 Bench Testing

Bench Testing was a success. The setup was a 3-channel power supply (NGE103), an electronic load (DSB3021), oscilloscope, laptop and the full system. Started at 200kHz open loop to get some baseline

behaviour. Then closed loop with gains of (0.1, 0.2, 0.3) of QSPICE values for pv panel setup. Then we progressed to 300 kHz at (0.3 and 0.5). Then to 400 kHz at 0.3. 300 kHz seemed to be the most optimal. There appears to be a trade-off where the benefits reduce as the switching speed increases and the heat increases. We also explored an open loop test on 100 kHz which the system has sort of working but bizarre behaviour to. It is concluded that the power supplies can not handle as much as the panels. Or do not require as aggressive control.
Limitations: Can only do 2 shufflers. Boards can not be stacked together. Can not comment on system scalability. The shufflers struggle to be driven independently as they are hardwired together. The top-level ina239 does not function had to guess based on eload value. Have to rerun the sim everytime the switching speed changes for best results. Did not test on solar panels so response to panel dynamics is only theoretical.

6.6 Solar Test

- solar testing was a success.
- raw peak sun data
- raw minimal sun data
- discuss the limitation of control loop testing on the pv was the weather. Not enough to give it a proper go.

6.7 Synthesising Results

- Input the comparison results.

7 Conclusion

Conclusions: summarise the key findings and any recommendations for further work

Say how accurate the control system is. How reliable it is. Significant it is to have such an elaborate system that can be used a bouncing off point for further investigations. Cycle back to the research question. Talk about how it absolutely smashed partial shading becuase my shirt was covering one of the panels and the contoller managed to create a massive energy return.

Future work: Scalability. Determine how the relationships change when more shufflers are acting in a string or how a shuffler handles varying number of panels. Streamline switching speed gain calcs. Generate and compare control models. Apply a machine learning model / cost function to determine optimal gains. Fix the top-level voltage sensor. Consider effectiveness with different panel arrangements. How does it stack up against dynamic panel configurations? Explore other applications outside the solar industry. Think about how it applies to other sectors of the energy industry like mining, computing, industrial, batteries etc. Would this let you win a solar car race? Is that game changing? Does the technology itself scale?

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Appendices

Risk Register

Likelihood and consequence are rated Low (L), Medium (M), High (H), or Critical (C). Overall risk rating is the higher of the two scores.

Table 2: Project Risk Register

#	Risk	Category	Likelihood	Consequence	Rating	Mitigation / Monitoring Strategy
1	Scope creep	Project	H	H	H	Strict change control; new requirements deferred to next design revision. Supervisor sign-off required for scope additions.
2	No PV hardware available	Resource	M	C	H	Procure panels early in Semester 1. Fallback: validate control system in QSpice HIL simulation and document as a known limitation.
3	Electronics failure	Hardware	M	H	H	Overload protection required by specification. Order spare components. One-month buffer absorbs a PCB reorder cycle.
4	Firmware failure	Control Sys.	M	H	H	MISRA-C enforced. QSpice HIL and MATLAB validate firmware before hardware deployment. Unit test plans with acceptance criteria.
5	Monitoring software inaccuracy	Software	L	M	L	Sensor noise filtering required by specification. Cross-validate logged data against known operating points during calibration.
6	Data loss	Data	L	H	M	GitHub for code; OneDrive auto-sync for datasets. Raw data backed up to secondary local drive after each session.
7	Unable to access data	Data	L	M	L	All assets in shared cloud repositories with at least two admin-level team members.
8	Electrical hazard (50 V DC)	Safety	M	C	**C**	QUT HSE electrical protocols enforced. PPE and isolation procedures mandatory. Formal risk assessment completed before any live test.
9	Sunburn (outdoor testing)	Safety	H	L	M	Sunscreen applied; outdoor sessions scheduled before 10 am or after 2 pm where practical.
10	Fire risk	Safety	L	C	H	Fire extinguisher present in lab. Rated fuses and current limits enforced. Energised hardware never left unattended.
11	Assignment deadline collision	Project	H	M	H	One-month contingency buffer in schedule. Milestones reviewed weekly. Non-critical tasks deprioritised during crunch periods.
12	Skill gap between team members	Team	M	M	M	Work division follows domain expertise (control vs. power electronics). Knowledge transfer at design review milestones.
13	Control–hardware integration failure	Control Sys.	M	H	H	QSpice HIL validates firmware–hardware interaction. Subsystem verification stage isolates faults before full system test.
14	MPPT fails to track global MPP	Control Sys.	M	H	H	Global MPP tracking is a hard requirement. Algorithm validated across all partial shading scenarios in simulation before deployment.
15	Intellectual property dispute	Legal	L	H	M	Monitor IP Australia. All design decisions timestamped in GitHub. Novel details withheld from publication prior to any IP filing.
16	Poor commercial adoption	Strategic	L	L	L	Outside capstone scope. Universal application requirement prevents overfitting to a single use case.
17	Data breach / cybersecurity	Data	L	M	L	No sensitive personal data collected. Repositories private. Credentials excluded from version control.
18	Team member illness	Team	M	M	M	Buffer absorbs short-term absence. Control systems work proceeds in isolation if hardware team is unavailable. Documentation kept current for handover.